

A MODEL SELECTION CRITERION FOR USE IN A DYNAMIC LINEAR MODELLING ENVIRONMENT

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Key words: AIC criterion; cross validation; Press residual; state space model.

Summary: A model selection criterion that is based on the principle of cross validation is developed for the state space model. A comparison with the more familiar AIC and Press criteria is then made through the medium of a practical example.

1. INTRODUCTION

The problem of model selection in a time series framework has been extensively studied in the literature. In particular, given an autoregressive process of unknown order k , Akaike (1969) has developed a final prediction error criterion to determine a suitable value for k . In a later study he developed, based on the principle of minimizing the Kullback-Leibler divergence measure, the following Akaike Information criterion:

$$AIC(k) = -2\ln(\text{maximum likelihood}) + 2k.$$

Given, however, the use of either of the above two selection criteria in an autoregressive modelling context, Shibata (1976) has shown that they will tend to overestimate the order of the autoregression. As a result, Schwartz (1978) developed a Bayesian information criterion which has been shown to be consistent in the sense that its use will result in the correct model structure being chosen with probability one as $n \rightarrow \infty$ (Azencott & Dacunha-Castelle (1976)). In 1974, Stone proposed a new method for model

selection that was based on the principle of splitting the available data into two parts, and then using the one part to derive a suitable prediction rule, and the other part to test the performance of this prediction rule. This technique of so-called cross validation, which was originally applied to a regression modelling framework, has subsequently been extended by Ansley and Kohn (1987, 1989) to a time series framework. In this paper we will make use of the ideas that were outlined in that paper to develop a cross validation technique that can be used in a dynamic linear modelling environment. Apart from the above reference, it would appear that the only other work that has been done in this area has been conducted by de Jong (1988) who developed a computationally exhaustive method for computing a cross validation filter that employs a recursive filtering technique.

2. A CROSS VALIDATION RESIDUAL FOR THE DYNAMIC LINEAR MODEL

Given a state space model specification of the form

$$\begin{aligned} \mathbf{y}_t &= \mathbf{F}_t \boldsymbol{\beta}_t + \mathbf{v}_t \\ \boldsymbol{\beta}_t &= \mathbf{G}_t \boldsymbol{\beta}_{t-1} + \mathbf{w}_t \\ \boldsymbol{\beta}_0 &\sim N(\hat{\boldsymbol{\beta}}_0|_0, \boldsymbol{\Sigma}_0|_0) \end{aligned}$$

where

$\mathbf{y}_t$ = an $m_t \times 1$ vector of process observations that have been collected at the time point t

$\boldsymbol{\beta}_t$ = a $p \times 1$ random vector of unknown regression coefficients (state variables)

$\mathbf{F}_t$ = an $m_t \times p$ matrix of regressor variables that are assumed to be known at the time point t

$\mathbf{G}_t$ = a $p \times p$ system equation matrix that is assumed to be known at the time point t

$\{\mathbf{v}_t\}$ = a sequence of uncorrelated normal random vectors with zero mean and variance covariance matrix $\mathcal{E}\{\mathbf{v}_t \mathbf{v}_t'\} = \mathbf{R}_t$

$\{w_t\}$ = a sequence of uncorrelated normal random vectors with zero mean and variance covariance matrix $E\{w_t w_t'\} = Q_t$

the purpose of this paper will be to develop a cross-validatory method for selecting the number of regressors to include in the matrix F_t . In particular, letting

$$\hat{\beta}_t|_n \equiv E\{\beta_t | y_1, \dots, y_n\}$$

$$\Sigma_t|_n \equiv E\{(\beta_t - \hat{\beta}_t|_t)(x_t - \hat{\beta}_t|_t)' | y_1, \dots, y_n\}.$$

Minimum mean square error filtering and smoothing estimates for the above model can be given by (Rauch, Tung & Striebel (1965), Shumway & Stoffer (1982))

$$\hat{\beta}_t|_t = \hat{\beta}_t|_{t-1} + K_t e_t$$

$$\Sigma_t|_t = \Sigma_t|_{t-1} - K_t F_t' \Sigma_t|_{t-1}$$

$$K_t = \Sigma_t|_{t-1} F_t' (F_t \Sigma_t|_{t-1} F_t' + R)^{-1}$$

$$\hat{\beta}_t|_{t-1} = G_t \hat{\beta}_{t-1}|_{t-1}$$

$$\Sigma_t|_{t-1} = G_t \Sigma_{t-1}|_{t-1} G_t' + Q_t$$

$$e_t = y_t - F_t \hat{\beta}_t|_{t-1}$$

$$\hat{\beta}_{t-1}|_n = \hat{\beta}_{t-1}|_{t-1} + J_{t-1} (\hat{\beta}_t|_n - G_t \hat{\beta}_{t-1}|_{t-1})$$

$$\Sigma_{t-1}|_n = \Sigma_{t-1} + J_{t-1} (\Sigma_t|_n - \Sigma_t|_{t-1}) J_{t-1}'$$

$$J_{t-1} = \Sigma_{t-1}|_{t-1} G_t' \Sigma_t^{-1}|_{t-1}$$

Adopting the notation

$$\mathbf{D}_n = \{\mathbf{y}_1, \dots, \mathbf{y}_n\}$$

$$\mathbf{D}_n^{(-t)} = \{\mathbf{y}_1 \dots \mathbf{y}_{t-1}, \mathbf{y}_{t+1} \dots \mathbf{y}_n\}$$

$$\hat{\beta}_t^{(-t)} = \mathcal{E}\{\beta_t | \mathbf{D}_n^{(-t)}\}$$

$$\Sigma_t^{(-t)} = \text{var}\{\beta_t | \mathbf{D}_n^{(-t)}\}$$

and letting $p(\beta_t | \mathbf{D}_n)$ denote the density function for β_t that is conditional on the data set $\mathbf{D}_n$, one can obtain the result that

$$p(\beta_t | \mathbf{D}_n) \propto p(\mathbf{y}_t | \mathbf{D}_n^{(-t)}, \beta_t) p(\mathbf{D}_n^{(-t)} | \beta_t) p(\beta_t)$$

$$\propto p(\mathbf{y}_t | \beta_t, \mathbf{D}_n^{(-t)}) p(\beta_t | \mathbf{D}_n^{(-t)})$$

which implies that

$$p(\beta_t | \mathbf{D}_n^{(-t)}) \propto p(\beta_t | \mathbf{D}_n) / p(\mathbf{y}_t | \beta_t, \mathbf{D}_n^{(-t)}).$$

Since

$$p(\mathbf{y}_t | \beta_t, \mathbf{D}_n^{(-t)}) \sim N(\mathbf{F}_t' \beta_t, \mathbf{R}_t)$$

and

$$p(\beta_t | \mathbf{D}_n) \sim N(\hat{\beta}_t | n, \Sigma_t | n)$$

it follows that

$$p(\beta_t | \mathbf{D}_n^{(-t)}) \propto \exp \left\{ -\frac{1}{2} [(\beta_t - \hat{\beta}_t | n)' \Sigma_t^{-1} | n (\beta_t - \hat{\beta}_t | n) - (\mathbf{y}_t - \mathbf{F}_t' \beta_t)' \mathbf{R}_t^{-1} (\mathbf{y}_t - \mathbf{F}_t' \beta_t)] \right\}.$$

Now because

$$(\mathbf{y}_t - \mathbf{F}_t' \beta_t)' \mathbf{R}_t^{-1} (\mathbf{y}_t - \mathbf{F}_t' \beta_t) = (\beta_t - \bar{\beta}_t)' \mathbf{F}_t' \mathbf{R}_t^{-1} \mathbf{F}_t (\beta_t - \bar{\beta}_t) + (\mathbf{y}_t - \mathbf{F}_t' \bar{\beta}_t)' \mathbf{R}_t^{-1} (\mathbf{y}_t - \mathbf{F}_t' \bar{\beta}_t)$$

where $\bar{\beta}_t$ satisfies

$$\mathbf{F}_t' \mathbf{R}_t^{-1} \mathbf{F}_t \bar{\boldsymbol{\beta}}_t = \mathbf{F}_t' \mathbf{R}_t^{-1} \mathbf{y}_t,$$

it follows that

$$\begin{aligned} p(\boldsymbol{\beta}_t | \mathbf{D}_n^{(-t)}) &\propto \exp \left\{ -\frac{1}{2} [(\boldsymbol{\beta}_t - \hat{\boldsymbol{\beta}}_{t|n})' \boldsymbol{\Sigma}_{t|n}^{-1} (\boldsymbol{\beta}_t - \hat{\boldsymbol{\beta}}_{t|n}) - (\boldsymbol{\beta}_t - \bar{\boldsymbol{\beta}}_t)' \mathbf{F}_t' \mathbf{R}_t^{-1} \mathbf{F}_t (\boldsymbol{\beta}_t - \bar{\boldsymbol{\beta}}_t)] \right\} \\ &\propto \exp \left\{ -\frac{1}{2} (\boldsymbol{\beta}_t - \hat{\boldsymbol{\beta}}_{t|n}^{(-t)})' \boldsymbol{\Sigma}_{t|n}^{(-t)} (\boldsymbol{\beta}_t - \hat{\boldsymbol{\beta}}_{t|n}^{(-t)}) \right\} \end{aligned}$$

where

$$\begin{aligned} \hat{\boldsymbol{\beta}}_{t|n}^{(-t)} &= (\boldsymbol{\Sigma}_{t|n}^{-1} - \mathbf{F}_t' \mathbf{R}_t^{-1} \mathbf{F}_t)^{-1} (\boldsymbol{\Sigma}_{t|n}^{-1} \hat{\boldsymbol{\beta}}_{t|n} - \mathbf{F}_t' \mathbf{R}_t^{-1} \mathbf{F}_t \bar{\boldsymbol{\beta}}_t) \\ &= (\boldsymbol{\Sigma}_{t|n}^{-1} - \mathbf{F}_t' \mathbf{R}_t^{-1} \mathbf{F}_t)^{-1} (\boldsymbol{\Sigma}_{t|n}^{-1} \hat{\boldsymbol{\beta}}_{t|n} - \mathbf{F}_t' \mathbf{R}_t^{-1} \mathbf{y}_t) \end{aligned}$$

and

$$\boldsymbol{\Sigma}_{t|n}^{(-t)} = (\boldsymbol{\Sigma}_{t|n}^{-1} - \mathbf{F}_t' \mathbf{R}_t^{-1} \mathbf{F}_t)^{-1}.$$

Making use of the following well-known matrix results for positive definite matrices

$$\begin{aligned} (\mathbf{A}^{-1} - \mathbf{B}' \mathbf{C}^{-1} \mathbf{B})^{-1} &= \mathbf{A} - \mathbf{A} \mathbf{B}' (\mathbf{B} \mathbf{A} \mathbf{B}' - \mathbf{C})^{-1} \mathbf{B} \mathbf{A} \\ (\mathbf{A}^{-1} - \mathbf{B}' \mathbf{C}^{-1} \mathbf{B})^{-1} \mathbf{B}' \mathbf{C}^{-1} &= \mathbf{A} \mathbf{B}' (\mathbf{C} - \mathbf{B} \mathbf{A} \mathbf{B}')^{-1} \\ \mathbf{B} (\mathbf{B}' \mathbf{C}^{-1} \mathbf{B} - \mathbf{A}^{-1}) \mathbf{B}' &= \mathbf{C} + \mathbf{C} (\mathbf{B} \mathbf{A} \mathbf{B}' - \mathbf{C})^{-1} \mathbf{C} \end{aligned}$$

one can obtain the following computing formulae for $\hat{\boldsymbol{\beta}}_{t|n}^{(-t)}$ and $\boldsymbol{\Sigma}_{t|n}^{(-t)}$, namely

$$\boldsymbol{\Sigma}_{t|n}^{(-t)} = (\boldsymbol{\Sigma}_{t|n}^{-1} - \mathbf{F}_t' \mathbf{R}_t^{-1} \mathbf{F}_t)^{-1} = \boldsymbol{\Sigma}_{t|n} - \boldsymbol{\Sigma}_{t|n} \mathbf{F}_t' (\mathbf{F}_t \boldsymbol{\Sigma}_{t|n} \mathbf{F}_t' - \mathbf{R}_t)^{-1} \mathbf{F}_t \boldsymbol{\Sigma}_{t|n}$$

and

$$\begin{aligned} \hat{\boldsymbol{\beta}}_{t|n}^{(-t)} &= (\boldsymbol{\Sigma}_{t|n} - \boldsymbol{\Sigma}_{t|n} \mathbf{F}_t' (\mathbf{F}_t \boldsymbol{\Sigma}_{t|n} \mathbf{F}_t' - \mathbf{R}_t)^{-1} \mathbf{F}_t \boldsymbol{\Sigma}_{t|n}) \boldsymbol{\Sigma}_{t|n}^{-1} \hat{\boldsymbol{\beta}}_{t|n} \\ &\quad - (\boldsymbol{\Sigma}_{t|n}^{-1} - \mathbf{F}_t' \mathbf{R}_t^{-1} \mathbf{F}_t)^{-1} \mathbf{F}_t' \mathbf{R}_t^{-1} \mathbf{y}_t \\ &= \hat{\boldsymbol{\beta}}_{t|n} - \boldsymbol{\Sigma}_{t|n} \mathbf{F}_t' (\mathbf{F}_t \boldsymbol{\Sigma}_{t|n} \mathbf{F}_t' - \mathbf{R}_t)^{-1} \mathbf{F}_t \hat{\boldsymbol{\beta}}_{t|n} - \boldsymbol{\Sigma}_{t|n} \mathbf{F}_t' (\mathbf{R}_t - \mathbf{F}_t \boldsymbol{\Sigma}_{t|n} \mathbf{F}_t')^{-1} \mathbf{y}_t \\ &= \hat{\boldsymbol{\beta}}_{t|n} - \boldsymbol{\Sigma}_{t|n} \mathbf{F}_t' (\mathbf{R}_t - \mathbf{F}_t \boldsymbol{\Sigma}_{t|n} \mathbf{F}_t')^{-1} (\mathbf{y}_t - \mathbf{F}_t \hat{\boldsymbol{\beta}}_{t|n}). \end{aligned}$$

On noting that

$$\begin{aligned}
 \mathcal{E}\{y_t | D_n^{(-t)}\} &= \mathcal{E}\{F_t \beta_t + v_t | D_n^{(-t)}\} \\
 &= F_t \hat{\beta}_{t|n}^{(-t)} \\
 &= F_t \hat{\beta}_{t|n} - F_t \Sigma_{t|n} F_t' (R_t - F_t' \Sigma_{t|n} F_t')^{-1} (y_t - F_t \hat{\beta}_{t|n}) \\
 \text{var}(y_t | D_n^{(-t)}) &= \text{var}\{F_t \beta_t + v_t | D_n^{(-t)}\} \\
 &= F_t \Sigma_{t|n}^{(-t)} F_t' + R_t \\
 &= F_t (\Sigma_{t|n}^{-1} - F_t' R_t^{-1} F_t)^{-1} F_t' + R_t \\
 &= F_t \Sigma_{t|n} F_t' + R_t - F_t \Sigma_{t|n} F_t' (F_t \Sigma_{t|n} F_t' - R_t)^{-1} F_t \Sigma_{t|n} F_t'
 \end{aligned}$$

a cross validation residual for the time point t can now be defined by

$$\begin{aligned}
 \tilde{e}_t &= y_t - \mathcal{E}\{y_t | D_n^{(-t)}\} \\
 &= y_t - F_t \hat{\beta}_{t|n}^{(-t)} \\
 &= y_t - F_t \hat{\beta}_{t|n} + F_t \Sigma_{t|n} F_t' (R_t - F_t' \Sigma_{t|n} F_t')^{-1} (y_t - F_t \hat{\beta}_{t|n}) \\
 &= \{I - F_t \Sigma_{t|n} F_t' (F_t \Sigma_{t|n} F_t' - R_t)^{-1}\} (y_t - F_t \hat{\beta}_{t|n})
 \end{aligned}$$

with the property that

$$\tilde{e}_t \sim (0, F_t \Sigma_{t|n}^{(-t)} F_t' + R_t).$$

3. APPLICATION AND CONCLUSION

To illustrate the use of the above cross-validatory technique, the following model selection problem was considered where an attempt was made to predict gasoline consumption in the United States as a function of an intercept term, and the following set of variables:

- x_1 = a price index for gasoline
- x_2 = a per capita real disposable income
- x_3 = a price index for new cars
- x_4 = a price index for used cars

The data which are given in table 1 were obtained from Greene (1990).

Subset dynamic linear model structures were then generated using the independent variables that are listed above. For example, given the subset model structure containing the variables x_1 and x_3 , a dynamic linear model of the form

$$\begin{aligned} y_t &= (1, x_1, x_3)\beta_t + v_t & v_t &\sim N(0, \sigma_r^2) \\ \beta_t &= \beta_{t-1} + w_t & w_t &\sim N(0, Q) \end{aligned}$$

was developed. The EM algorithm of Dempster, Laird and Rubin (1976) was used to estimate the unknown noise variance covariance parameters, σ_r^2 and Q , and then, using the notation of the previous section, the following cross-validatory criterion was computed

$$CV = \sum_{t=1}^n \tilde{e}_t' (F_t \Sigma_t^{(-t)} F_t' + R_t)^{-1} \tilde{e}_t$$

The results of such an analysis, for all possible subset regression model structures, is given in table 2.

For the purposes of making a comparison with a constant parameter linear regression model structure, a corresponding Press and AIC model selection value, for each of the subset model structures was also computed.

If one were to assume that a more traditional constant parameter regression model holds true, then an examination of the results in table 2 would indicate that the subset model structure $\{x_1, x_2, x_3\}$ should, from a Press point of view, be preferred. The MSE-based criterion also confirms this where the model structure $\{x_1, x_2, x_3\}$ would once again be marginally preferred to that of $\{x_1, x_2\}$. Employment of the AIC model selection

criterion would, however, lead to the selection of the subset model structure $\{x_2, x_3, x_4\}$. If, however, one were to make the assumption that the regression coefficient parameters follow a first-order Markov process of the form that is given in our dynamic linear model, then the model structure $\{x_1, x_3, x_4\}$ would be preferred.

As an additional example which serves primarily to illustrate the use of our cross validation criterion in a non-nested model selection environment, consider the problem of trying to estimate the quarterly number of car unit sales in the United States as a function of an intercept term (x_0), and the following variables:

x_1 = an index reflecting consumer sentiment (1966:I=100)

x_2 = the average miles travelled per cost of car ownership
(1972=1,000)

x_3 = the average miles travelled per gallon for new cars

x_4 = the loan finance rate on a new car

d_1 = a seasonal dummy index for the first quarter

d_2 = a seasonal dummy index for the second quarter

d_3 = a seasonal dummy index for the third quarter

The data which are listed in table 3 were obtained from Bails and Peppers (1982).

Employing some suitably chosen transformation techniques on both the dependent and independent variables that are listed in table 3, the following four competing dynamic linear model structures were calculated, viz:

Model I: $\{y, x_0, x_1, x_2, x_3, x_4, d_1, d_2, d_3\}$

Model II: $\{\ln(y), x_0, x_1, x_2, x_3, x_4, d_1, d_2, d_3\}$

Model III: $\{\ln(y), x_0, x_1, x_2, x_3, x_4\}$

Model IV: $\{\ln(y), x_0, \ln(x_1), \ln(x_2), \ln(x_3), \ln(x_4)\}$

An examination of the results that are listed in table 4 would indicate that among those model structures involving the logarithmic transformation, the model structure III would be preferred from a cross-validatory point of view.

Apart from illustrating the practical implementation of our cross-validatory model selection technique, the above examples serve to highlight the effect that the introduction of a time-varying regression coefficient parameter may have on the model selection process. Clearly, this and other problems regarding the use of the above model selection technique need to be examined further. In particular, the cross validation residuals that have been derived in this paper have been based on the principle of leaving one observation out at a time. Efron (1986), however, has shown that such a technique is prone to overfitting. As a result Breiman and Spector (1989) and Burman (1990) have suggested that one delete $d > 1$ observations at a time, providing simulation evidence that such a technique is more preferable.

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Table 1. Gasoline consumption data

Date	Y	X_1	X_2	X_3	X_4
1960	129,7	0,925	6036	1,045	0,836
1961	131,3	0,914	6113	1,045	0,869
1962	137,1	0,919	6271	0,948	0,948
1963	141,6	0,918	6378	1,035	0,960
1964	148,8	0,914	6727	1,032	1,001
1965	155,9	0,949	7027	1,009	0,994
1966	164,9	0,970	7280	0,991	0,970
1967	171,0	1,000	7513	1,000	1,000
1968	183,4	1,014	7728	1,028	1,028
1969	195,8	1,047	7891	1,044	1,031
1970	207,4	1,056	8134	1,076	1,043
1971	218,3	1,063	8322	1,120	1,102
1972	226,8	1,076	8562	1,110	1,105
1973	237,9	1,181	9042	1,111	1,176
1974	225,8	1,599	8867	1,175	1,226
1975	232,4	1,708	8944	1,276	1,464
1976	241,7	1,779	9175	1,357	1,679
1977	249,2	1,882	9381	1,429	1,828
1978	261,3	1,963	9735	1,538	1,865
1979	248,9	2,656	9829	1,660	2,010
1980	226,8	3,691	9722	1,793	2,081
1981	225,6	4,109	9769	1,902	2,569
1982	228,8	3,894	9725	1,976	2,964
1983	239,6	3,764	9930	2,026	3,297
1984	244,7	3,707	10421	2,085	3,757
1985	245,8	3,738	10563	2,152	3,797
1986	269,4	2,921	10780	2,240	3,632

Source: Greene (1990, p. 206)

Table 2. Model selection results for the gasoline consumption data

Variables	Press	MSE	CV	σ_r^2	AIC
none	87,5128	1,4342	24,5666	0,1770	35,661
x_1	91,3905	1,4589	24,8288	0,1860	36,523
x_2	73,7991	1,1994	23,2768	0,2329	43,809
x_3	77,9435	1,2691	25,2864	0,1567	33,044
x_4	84,1108	1,3677	22,1358	0,2221	40,413
$x_1 x_2$	73,2484	1,1523	23,4533	0,1787	60,001
$x_1 x_3$	81,6589	1,2888	22,7917	0,2163	24,741
$x_1 x_4$	80,4845	1,2750	23,3669	0,0875	39,436
$x_2 x_3$	75,1259	1,2131	24,1535	0,2089	45,485
$x_2 x_4$	74,4124	1,1918	22,2192	0,1809	38,607
$x_3 x_4$	78,6163	1,2589	26,1567	0,1048	31,985
$x_1 x_2 x_3$	73,2366	1,1416	22,0728	0,1806	31,130
$x_1 x_2 x_4$	75,2478	1,1724	21,9162	0,0720	30,198
$x_1 x_3 x_4$	80,8430	1,2518	21,0863	0,0610	26,059
$x_2 x_3 x_4$	75,2468	1,2108	21,3659	0,0544	12,001
$x_1 x_2 x_3 x_4$	74,4885	1,1601	22,3490	0,0484	12,454

Table 3. Quarterly car sales data

Date	Y	X ₁	X ₂	X ₃	X ₄
1965: I	9,20	102	0,782	14,1	8,09
II	8,60	102	0,780	14,1	8,02
III	8,67	103	0,778	14,1	8,02
IV	8,67	103	0,778	13,9	8,10
1966: I	9,87	100	0,783	13,9	8,21
II	8,57	96	0,795	13,9	8,50
III	8,87	91	0,812	13,9	8,88
IV	8,90	88	0,827	13,5	9,13
1967: I	8,00	92	0,817	13,5	8,9
II	8,90	95	0,816	13,5	8,69
III	8,50	97	0,830	13,5	8,73
IV	8,07	93	0,846	13,6	8,89
1968: I	9,40	95	0,862	13,6	9,08
II	9,37	92	0,876	13,6	9,40
III	9,97	93	0,895	13,6	9,81
IV	9,80	92	0,900	13,5	9,77
1969: I	9,67	95	0,916	13,5	10,03
II	9,60	92	0,940	13,5	10,41
III	9,60	86	0,958	13,5	10,86
IV	9,40	80	0,972	13,0	11,06
1970: I	8,77	78	0,984	13,0	11,34
II	9,13	77	1,001	13,0	11,30
III	9,00	77	1,001	13,0	11,37
IV	6,87	75	1,024	12,6	11,25
1971: I	10,03	78	1,025	12,6	10,59
II	9,97	82	0,999	12,6	10,12
III	10,40	82	1,005	12,6	10,42
IV	10,63	82	0,996	12,6	10,41
1972: I	10,53	88	1,001	12,6	10,20
II	10,77	89	0,995	12,6	9,96
III	10,93	94	1,004	12,6	10,02
IV	11,63	91	1,000	13,3	10,02
1973: I	12,37	81	1,009	13,3	10,05
II	12,23	76	1,023	13,3	10,05
III	11,20	72	1,042	13,3	10,25
IV	9,93	76	1,080	12,1	10,49
1974: I	9,43	61	1,131	12,1	10,53
II	9,27	72	1,194	12,1	10,65
III	9,63	64	1,261	12,1	11,14
IV	7,53	58	1,310	13,6	11,57
1975: I	8,33	58	1,315	13,6	11,53
II	7,85	73	1,336	13,6	11,36
III	9,06	76	1,369	13,6	11,31
IV	9,38	75	1,389	14,3	11,24
1976: I	10,17	85	1,396	14,3	11,17
II	10,08	82	1,407	14,3	11,04
III	10,08	82	1,407	14,3	11,04
IV	10,24	86	1,470	15,3	11,03
1977: I	11,28	88	1,492	15,3	11,03
II	11,54	89	1,489	15,3	10,82
III	10,93	88	1,498	15,3	10,85
IV	10,98	83	1,522	15,8	10,87
1978: I	10,82	78	1,550	15,8	10,86
II	12,04	84	1,571	15,8	10,84
III	11,23	79	1,629	15,8	11,09
IV	11,13	82	1,666	16,4	11,29
1979: I	11,63	83	1,745	16,4	11,60
II	10,53	80	1,854	16,4	11,73
III	10,77	82	1,958	16,4	11,88
IV	9,83	78	2,083	17,0	12,85

Source: Bails & Peppers (1982, p. 449)

Table 4. Model selection results for the car sales data

Model	Press	MSE	CV	σ_r^2
I	74,4885	1,1602	27,1191	0,0950
II	0,8494	0,0125	0,3728	0,0002
III	0,7920	0,0123	0,3547	0,0008
IV	0,7045	0,0111	0,3632	0,0008